

Workshop on Advanced Sampling Methodologies

Google Earth Engine (GEE) Code Editor tutorial

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Open the Code Editor

- Go to <https://code.earthengine.google.com/>.
- Make sure you're signed in with a Google account that has Earth Engine access.

Load a Shapefile

There are two ways:

Option A: Upload a shapefile

1. In the **Assets** tab (left panel), click **NEW → Table upload**.
2. Upload your **.shp** (plus **.shx**, **.dbf**, etc. in a zipped folder).
3. Once uploaded, it appears under your assets.

Option B: Use an existing dataset

```
1 // Example: Use a country boundary from FAO GAUL dataset
2 var region = ee.FeatureCollection("FAO/GAUL/2015/level0")
3           .filter(ee.Filter.eq('ADM0_NAME', 'Kenya'));
4 Map.centerObject(region, 6);
```

Load an image and subset to shapefile

Example with **Sentinel-2** surface reflectance:

```
1 // Load Sentinel-2 surface reflectance
2 var image = ee.ImageCollection("COPERNICUS/S2_SR")
3     .filterDate('2021-01-01', '2021-12-31')
4     .filterBounds(region)
5     .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 10))
6     .median();
7
8 // Clip the image to shapefile boundaries
9 var clipped = image.clip(region);
```

Explore available bands

Check which bands are included:

```
1 print('Sentinel-2 bands:', image.bandNames());
```

Example band names: B2 (blue), B3 (green), B4 (red), B8 (NIR).

Visualize the image

```
1 // True color (RGB) visualization
2 var visParams = {
3   bands: ['B4', 'B3', 'B2'],
4   min: 0,
5   max: 3000,
6 };
7
8 Map.addLayer(clipped, visParams, 'Sentinel-2 RGB');
9
10 // False color (NIR, Red, Green)
11 var falseColor = {
12   bands: ['B8', 'B4', 'B3'],
13   min: 0,
14   max: 3000,
15 };
16 Map.addLayer(clipped, falseColor, 'False Color');
```

Export the image

Export to **Google Drive**:

```
1 Export.image.toDrive({  
2   image: clipped,  
3   description: 'Sentinel2_Kenya_2021',  
4   folder: 'EarthEngineExports',  
5   scale: 10, // Sentinel-2 resolution  
6   region: region.geometry(),  
7   maxPixels: 1e13  
8 });
```

Export to **Asset** (for later use in GEE):

```
1 Export.image.toAsset({  
2   image: clipped,  
3   description: 'Sentinel2_Kenya_Asset',  
4   assetId: 'users/your_username/Sentinel2_Kenya',  
5   scale: 10,  
6   region: region.geometry(),  
7   maxPixels: 1e13  
8 });
```

Summary

- Upload/load a shapefile as a `FeatureCollection`.
- Filter an image collection and clip it to your shapefile.
- Print and inspect band names.
- Visualize with `Map.addLayer()` using different bands.
- Export results to Drive or Assets.